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Bihar Engineering University, Patna

B.Tech 4th Semester Exam., 2025

Code: 101403 ***Time: 03 Hours***

Semester: IV ***Subject: Engineering Geology*** ***Full Marks: 70***

Instructions:-

(i) The marks are indicated in the right-hand margin.
(ii) There are NINE questions in this paper.
(iii) Attempt FIVE questions in all.
(iv) Question No. 1 is compulsory.

Q.1 Choose the correct option / answer the following (Any seven question only): **[2 x 7 = 14]**

(a) A solid barrier constructed at a suitable location across a river valley to impound water is called
(i) Reservoir (iii) Bridge
(ii) Dam (iv) Retaining wall

(b) Which type of dam usually has a triangular profile and can resist the forces by its own weight?
(i) Gravity dam (iii) Geotechnical dam
(ii) Arch dam (iv) Embankment dam

(c) Which type of rocks is classified based on geometry?
(i) Pseudo-stratified (iii) Non-stratified
(ii) Stratified (iv) Anisotropic

(d) Joints parallel to bedding planes are called
(i) Strike joints (iii) Dip joints
(ii) Bedding joints (iv) Oblique joints

(e) Example for elongated structure is
(i) Calcite (iii) Orthoclase
(ii) Beryl (iv) Barite

(f) The water which is in a state of downward movement under gravity is
(i) Groundwater (iii) Connate water
(ii) Vadose water (iv) Juvenile water

(g) What is the process of movement downwards of vadose water called?
(i) Infiltration (iii) Deposition
(ii) Filtration (iv) Down-flow

(h) The branch or study which deals with behaviour of rocks under applied forces is called
(i) Soil mechanics (iii) Lithology
(ii) Rock mechanics (iv) Geology

(i) Which rock possesses very high compressive strength
(i) Igneous (iii) Metamorphic
(ii) Sedimentary (iv) Sedimentary and metamorphic

(j) The rock-mineral insoluble in water is
(i) Rock salt (iii) Calcite
(ii) Gypsum (iv) Pyrite

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Q.2 Write the answers in short: **[7 x 2 = 14]**

(a) Describe the different types of rocks. Give the classification, texture and structure of igneous, sedimentary and metamorphic rocks.

(b) On the basis of silicate structure, classify silicate minerals into various groups. Explain the structure of each group in detail.

Q.3 Write the answers in short: **[7 x 2 = 14]**

(a) Illustrate the various types of folds with neat sketches.

(b) How is Geology related to Engineering? Discuss the scope of application of geological knowledge in the planning work?

Q.4 Write the answers in short: **[7 x 2 = 14]**

(a) Enumerate with appropriate figures on the types, causes of landslides and their mitigation measures.

(b) Explain the classification of sedimentary rocks giving suitable example?

Q.5 Write the answers in short: **[7 x 2 = 14]**

(a) Classify and describe the different types of faults? Give the various minor structures found in the fault Zones? Discuss the effects of faulting on various engineering projects?

(b) Describe in detail various components of the hydrological cycle. Add a note on the factors controlling the movement of ground water?

Q.6 Write the answers in short: **[7 x 2 = 14]**

(a) Explain the classification of sedimentary rocks giving suitable example?

(b) Distinguish between weathering and erosion. Describe the various process of weathering.

Q.7 Write the answers in short: **[7 x 2 = 14]**

(a) Discuss Geological condition required for selection of dam and reservoirs site.

(b) Describe the various effects of an earthquake? How do we locate the epicentre of an earth quake?

Q.8 Write the answers in short: **[7 x 2 = 14]**

(a) What are Classification of Igneous rocks on the basis of Chemical composition?

(b) How do civil engineers determine the extent of weathering pattern in major civil engineering constructions?

Q.9 Write short note on any two of the following: **[7 x 2 = 14]**

(a) Retaining wall

(b) Rock bolting

(c) Failure theories and shear strength of rocks

(d) Bearing capacity of rocks

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